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Effective Teaching in Elementary Mathematics:

Identifying Classroom Practices that Support Student Achievement

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Abstract

Recent investigations into the education production function have moved beyond traditional teacher inputs, such as education, certification, and salary, focusing instead on observational measures of teaching practice. However, challenges to identification mean that this work has yet to coalesce around specific instructional dimensions that increase student achievement. I build on this discussion by exploiting within-school, between-grade, and cross-cohort variation in scores from two observation instruments; further, I condition on a uniquely rich set of teacher characteristics, practices, and skills. Findings indicate that inquiry-oriented instruction positively predicts student achievement. Content errors and imprecisions are negatively related, though these estimates are sensitive to the set of covariates included in the model. Two other dimensions of instruction, classroom emotional support and classroom organization, are not related to this outcome. Findings can inform recruitment and development efforts aimed at improving the quality of the teacher workforce.

JEL Classifications: Analysis of Education (I21), Human Capital (J24), Econometrics (C01)

Highlights

- This paper examines the relationship between high-quality instruction and students' mathematics achievement.
- Identification relies on variation in teaching quality within schools, across grades and years, conditioning on a unique set of teacher characteristics.
- Inquiry-oriented mathematics instruction is positively related to student outcomes with an effect size of 0.10 standard deviations.
- Findings generalize to teachers and classrooms across three urban school districts.

1. Introduction

Over the past decade, research has confirmed that teachers have substantial impacts on their students' academic and life-long success (e.g., Nye, Konstantopoulos, & Hedges, 2004; Chetty, Friedman, & Rockoff, 2014). Despite concerted efforts to identify characteristics such as experience, education, and certification that might be correlated with effectiveness (for a review, see Wayne & Youngs, 2003), however, the nature of effective teaching still largely remains a black box. Given that the effect of teachers on achievement must occur at least in part through instruction, it is critical that researchers identify the types of classroom practices that matter most to student outcomes. This is especially true as schools and districts work to meet the more rigorous goals for student achievement set by the Common Core State Standards (Porter, McMaken, Hwang, & Yang, 2011), particularly in mathematics (Duncan, 2010; Johnson, 2012; U.S Department of Education, 2010).

Our limited progress toward understanding the impact of teaching practice on student outcomes stems from two main research challenges. The first barrier is developing appropriate tools to measure the quality of teachers' instruction. Much of the work in this area tends to examine instruction either in laboratory settings or in classrooms over short periods of time (e.g., Anderson, Everston, & Brophy, 1979; Star & Rittle-Johnson, 2009), neither of which is likely to capture the most important kinds of variation in teachers' practices that occur over the course of a school year. The second is a persistent issue in economics of education research of designing studies that support causal inferences (Murnane & Willett, 2011). Non-random sorting of students to teachers (Clotfelter, Ladd, & Vigdor, 2006; Rothstein, 2010) and omitted measures of teachers' skills and practices limit the success of prior research.

I address these challenges through use of a unique dataset on fourth- and fifth-grade teachers and their students from three anonymous school districts on the East Coast of the United States. Over the course of two school years, the project captured observed measures of teachers' classroom practices on the Mathematical Quality of Instruction (MQI) and Classroom Assessment Scoring System (CLASS) instruments, focusing on mathematics-specific and general teaching practices, respectively. The project also collected data on a range of other teacher characteristics, as well as student outcomes on a low-stakes achievement test that was common across participants.

My identification strategy has two key features that distinguish it from prior work on this topic. First, to account for sorting of students to schools and teachers, I exploit variation in observation scores within schools, across adjacent grades and years. Specifically, I specify models that include school fixed effects and instructional quality scores averaged to the school-grade-year level. This approach assumes that student and teacher assignments are random within schools and across grades or years, which I explore in detail below. Second, to isolate the independent contribution of instructional practices to student achievement, I condition on a uniquely rich set of teacher characteristics, skills, and practices. I expect that there likely are additional factors that are difficult to observe and, thus, are excluded from my data. Therefore, to explore the possible degree of bias in my estimates, I test the sensitivity of results to models that include different sets of covariates. Further, I interpret findings in light of limitations associated with this approach.

Results point to a positive relationship between ambitious or inquiry-oriented mathematics instruction and performance on a low-stakes test of students' math knowledge of roughly 0.10 standard deviations. I also find suggestive evidence for a negative relationship

between mathematical errors and student achievement, though estimates are sensitive to the specific set of teacher characteristics included in the model. I find no relationships between two other dimensions of teaching practice – classroom emotional support and classroom organization – and student achievement. Teachers included in this study have value-added scores calculated from state assessment data similar to those of other fourth- and fifth-grade teachers in their respective districts, leading me to conclude that findings likely generalize to these populations beyond my identification sample. I argue that results can inform recruitment and development efforts aimed at improving the quality of the teacher workforce

The remainder of this paper is organized as follows. In the second section, I discuss previous research on the relationship between observational measures of teacher quality and student achievement. In the third section, I describe the research design, including the sample and data. In the fourth section, I present my identification strategy and tests of assumptions. In the fifth section, I provide main results and threats to internal and external validity. I conclude by discussing the implications of my findings for ongoing research and policy on teacher and teaching quality.

2. Background and Context

Although improving the quality of the teacher workforce is seen as an economic imperative (Hanushek, 2009), long-standing traditions that reward education and training or offer financial incentives based on student achievement have been met with limited success (Boyd, Grossman, Lankford, Loeb, & Wyckoff, 2006; Fryer, 2013; Harris & Sass, 2011; Springer et al., 2010). One reason for this posed by Murnane and Cohen (1986) almost three decades ago is the "nature of teachers' work" (p. 3). They argued that the "imprecise nature of the activity" makes it

difficult to describe *why* some teachers are good and what other teachers can do to improve (p. 7).

Recent investigations have sought to test this theory by comparing subjective and objective (i.e., "value-added") measures of teacher performance. In one such study, Jacob and Lefgren (2008) found that principals were able to distinguish between teachers in the tails of the achievement distribution but not in the middle. Correlations between principal ratings of teacher effectiveness and value added were weak to moderate: 0.25 and 0.18 in math and reading, respectively (0.32 and 0.29 when adjusted for measurement error). Further, while subjective ratings were a statistically significantly predictor of future student achievement, they performed worse than objective measures. Including both in the same regression model, estimates for principal ratings were 0.08 standard deviations (sd) in math and 0.05 sd in reading; comparatively, estimates for value-added scores were 0.18 sd in math and 0.10 sd in reading. This evidence led the authors to conclude that "good teaching is, at least to some extent, observable by those close to the education process even though it may not be easily captured in those variables commonly available to the econometrician" (p. 103).

Two other studies found similar results. Using data from New York City, Rockoff, Staiger, Kane, and Taylor (2012) estimated correlations of roughly 0.21 between principal evaluations of teacher effectiveness and value-added scores averaged across math and reading. These relationships corresponded to effect sizes of 0.07 sd in math and 0.08 sd in reading when predicting future student achievement. Extending this work to mentor evaluations of teacher effectiveness, Rockoff and Speroni (2010) found smaller relationships to future student achievement in math between 0.02 sd and 0.05 sd. Together, these studies suggest that principals

and other outside observers understand some but not all of the production function that converts classroom teaching and professional expertise into student outcomes.

In more recent years, there has been a growing interest amongst educators and economists alike in exploring teaching practice more directly. This now is possible through the use of observation instruments that quantitatively capture the nature and quality of teachers' instruction. In one of the first econometric analyses of this kind, Kane, Taylor, Tyler, and Wooten (2011) examined teaching quality scores captured on the Framework for Teaching instrument as a predictor of math and reading test scores. Data came from Cincinnati and widespread use of this instrument in a peer evaluation system. Relationships to student achievement of 0.11 sd in math and 0.14 sd in reading provided suggestive evidence of the importance of general classroom practices captured on this instrument (e.g., classroom climate, organization, routines) in explaining teacher productivity.

At the same time, this work highlighted a central challenge associated with looking at relationships between scores from observation instruments and student test scores. Non-random sorting of students to teachers and non-random variation in classroom practices across teachers means that there likely are unobserved characteristics related both to instructional quality and student achievement. As one way to address this concern, the authors' preferred model included school fixed effects to account for factors at the school level, apart from instructional quality, that could lead to differences in achievement gains. In addition, they relied on out-of-year observation scores that, by design, could not be correlated with the error term predicting current student achievement. This approach is similar to those taken by Jacob and Lefgren (2008), Rockoff, Staiger, Kane, and Taylor (2012), and Rockoff and Speroni (2010), who use principal/mentor ratings of teacher effectiveness to predict future student achievement. Finally,

as a robustness test, the authors fit models with teacher fixed effects to account for time-invariant teacher characteristics that might be related to observation scores and student outcomes; however, they noted that these estimates were much noisier because of small samples of teachers.

The largest and most ambitious study to date to conduct these sorts of analyses is the Measures of Effective Teaching (MET) project, which collected data from teachers across six urban school districts on multiple observation instruments. By randomly assigning teachers to class rosters within schools and using out-of-year observation scores, Kane, McCaffrey, Miller, and Staiger (2013) were able to limit some of the sources of bias described above. In math, relationships between scores from the Framework for Teaching and prior student achievement fell between 0.09 sd and 0.11 sd. In the non-random assignment portion of the study, Kane and Staiger (2012) found correlations between scores from other observation instruments and prior-year achievement gains in math from 0.09 (for the Mathematical Quality of Instruction) to 0.27 (for the UTeach Teacher Observation Protocol). The authors did not report these as effect size estimates. As a point of comparison, the correlation for the Framework for Teaching and prior-year gains was 0.13.

Notably, these relationships between observation scores and student achievement from both the Cincinnati and MET studies are equal to or larger in magnitude than those that focus on principal or mentor ratings of teacher quality. This is somewhat surprising given that principal ratings of teacher effectiveness – often worded specifically as teachers' ability to raise student achievement – and actual student achievement are meant to measure the same underlying construct. Comparatively, dimensions of teaching quality included on these instruments are thought to be important contributors to student outcomes but are not meant to capture every

aspect of the classroom environment that influence learning (Pianta & Hamre, 2009). Therefore, using findings from Jacob and Lefgren (2008), Rockoff, Staiger, Kane, and Taylor (2012), and Rockoff and Speroni (2010) as a benchmark, estimates describing the relationship between observed classroom practices and student achievement are, at a minimum, substantively meaningful; at a maximum, they may be viewed as large. Following Murnane and Cohen's intuition, then, continued exploration into the "nature of teachers' work" (1986, p. 3), the practices that comprise high-quality teaching, and their role in the education production function will be a central component of efforts aimed at raising teacher quality and student achievement.

At the same time that work by Kane and his co-authors (2011, 2012, 2013) has greatly expanded conversation in the economics of education literature to include teaching quality when considering teacher quality, this work has yet to coalesce around specific instructional dimensions that increase student outcomes. Random assignment of teachers to students – and other econometric methods such as use of school fixed effects, teacher fixed effects, and out-of-year observation ratings – likely provide internally valid estimates of the *effect of having a teacher who provides high-quality instruction* on student outcomes. This approach is useful when validating different measures of teacher quality, as was the stated goal of many of the studies described above including MET. However, these approaches are insufficient to produce internally valid estimates of the *effect of high-quality instruction* itself on student outcomes. This is because teachers whose measured instructional practices are high quality might have a true, positive effect on student achievement even though other practices and skills – e.g., spending more time with students, knowledge of students – are responsible for the higher achievement. Kane et al. (2011) fit models with teacher fixed effects in order to "control for all time-invariant teacher characteristics that might be correlated with both student achievement growth and

observed classroom practices" (p. 549). However, it is likely that there are other time-variant skills related both to instructional quality and student achievement.

I address this challenge to identification in two ways. First, my analyses explore an additional approach to account for the non-random sorting of students to teachers. Second, I attempt to isolate the unique contribution of specific teaching dimensions to student outcomes by conditioning on a broad set of teacher characteristics, practices, and skills. Specifically, I include observation scores captured on two instruments (both content-specific and general dimensions of instruction), background characteristics (education, certification, and teaching experience), knowledge (mathematical content knowledge and knowledge of student performance), and non-instructional classroom behaviors (preparation for class and formative assessment) that are thought to relate both to instructional quality and student achievement. Comparatively, in their preferred model, Kane et al. (2011) included scores from one observation instrument, controlling for teaching experience. While I am not able to capture every possible characteristic, I argue that these analyses are an important advance beyond what currently exists in the field.

3. Sample and Data

3.1 Sample

Data come from the National Center for Teacher Effectiveness (NCTE), which focused on collection of instructional quality scores and other teacher characteristics in three anonymous districts (henceforth Districts 1 through 3). Districts 1 and 2 are located in the same state. Data was collected from participating fourth- and fifth-grade math teachers in the 2010-2011 and 2011-2012 school years. Due to the nature of the study and the requirement for teachers to be

This project also includes a fourth district that I exclude here due to data and sample limitations. In the first year of the study, students did not take the baseline achievement test. In the second year, there were only three schools in which all teachers in the relevant grades participated in data collection, which is an important requirement of my identification strategy. At the same time, when I include these few observations in my analyses, patterns of results are the same.

videotaped over the course of a school year, participants consist of a non-random sample of schools and teachers who agreed to participate. During recruitment, study information was presented to schools based on district referrals and size; the study required a minimum of two teachers at each of the sampled grades. Of eligible teachers, 143 (roughly 55%) agreed to participate. My identification strategy focuses on school-grade-years in which I have the full sample of teachers who work in non-specialized classrooms (i.e., not self-contained special education or limited English proficient classes) in that school-grade-year. I further restrict the sample to schools that have at least two complete grade-year cells. This includes 111 teachers in 26 schools and 76 school-grade-years; 45 of these teachers, 17 of these schools, and 27 of these school-grade-years are in the sample for both school years.

In Table 1, I present descriptive statistics on the students and teachers in this sample. Students in District 1 are predominantly African American or Hispanic, with over 80% eligible for free- or reduced-price lunch (FRPL), 15% designated as in need of special education (SPED) services, and roughly 24% designated as limited English proficient (LEP). In District 2, there is a greater percentage of white students (29%) and fewer FRPL (71%), SPED (10%), and LEP students (18%). In District 3, there is a greater percentage of African-American students (67%) and fewer FRPL (58%), SPED (8%), and LEP students (7%). Across all districts, teachers have roughly nine years of experience. Teachers in Districts 1 and 2 were certified predominantly through traditional programs (74% and 93%, respectively), while more teachers in District 3 entered the profession through alternative programs or were not certified at all (55%). Relative to all study participants, teachers in Districts 1 through 3 have above average, average, and below average mathematical content knowledge, respectively.

Table 1
Sample Descriptive Statistics

	All Districts	District 1	District 2	District 3
Students				
Male (%)	49.7	48.8	51.1	47.6
African American (%)	53.1	42.8	51.0	67.2
Asian (%)	4.2	7.2	3.7	2.4
Hispanic (%)	17.2	37.7	12.4	8.8
White (%)	21.7	6.6	29.0	19.8
FRPL (%)	71.0	84.1	71.3	58.3
SPED (%)	10.6	14.5	10.2	7.9
LEP (%)	16.4	23.6	17.8	6.6
Students	3203	724	1692	787
Teachers				
Bachelor's Degree in Education (%)	45.4	33.3	57.5	42.1
Math Coursework (Likert Scale from 1 to 4)	2.3	2.4	2.4	2.2
Master's Degree (%)	75.0	83.3	77.5	65.8
Traditional Certification (%)	70.3	74.2	92.5	45.0
Experience (In Years)	9.0	8.9	9.1	9.0
Mathematical Content Knowledge (Standardized)	-0.07	0.15	0.00	-0.35
Knowledge of Student Performance (Standardized)	0.05	0.32	0.16	-0.28
Preparation for Class (Likert Scale from 1 to 5)	3.4	3.4	3.3	3.4
Formative Assessment (Likert Scale from 1 to 5)	3.6	3.6	3.6	3.6
Teachers	111	31	40	40

3.2 Main Predictor and Outcome Measures

3.2.1 Video-Recorded Lesson of Instruction

Mathematics lessons were captured over a two-year period, with a maximum of three lessons per teacher per year. Capture occurred with a three-camera, unmanned unit and lasted between 45 and 80 minutes. Teachers were allowed to choose the dates for capture in advance, and were directed to select typical lessons and exclude days on which students were taking a test. Although it is possible that these lessons are unique from teachers' general instruction, teachers did not have any incentive to select lessons strategically as no rewards or sanctions were involved with data collection. In addition, analyses from the MET project indicate that teachers are ranked almost identically when they choose lessons themselves compared to when lessons are chosen for them (Ho & Kane, 2013).

Trained raters scored these lessons on two established observational instruments: the Mathematical Quality of Instruction (MQI), focused on mathematics-specific practices, and the Classroom Assessment Scoring System (CLASS), focused on general teaching practices. For the MQI, two certified and trained raters watched each lesson and scored teachers' instruction on 17 items for each seven-and-a-half minute segment on a scale from Low (1) to High (3) (see Table 2 for a full list of items). Lessons have different numbers of segments, depending on their length. Analyses of these data (Blazar, Braslow, Charalambous, & Hill, 2015) show that items cluster into two main factors: *Ambitious Mathematics Instruction*, which corresponds to many elements contained within the mathematics reforms of the 1990s (National Council of Teachers of Mathematics, 1989, 1991, 2000) and the *Common Core State Standards for Mathematics* (National Governors Association for Best Practices, 2010); and *Mathematical Errors and Imprecisions,* which captures any mathematical errors the teacher introduces into the lesson. For *Ambitious Mathematics Instruction*, higher scores indicate better performance. For *Mathematical Errors and Imprecisions*, higher scores indicate that teachers make more errors in their instruction and, therefore, worse performance. I estimate reliability for these metrics by calculating the amount of variance in teacher scores that is attributable to the teacher (i.e., the intraclass correlation), adjusted for the modal number of lessons. These estimates are 0.69 and 0.52 for *Ambitious Mathematics Instruction* and *Mathematical Errors and Imprecisions*, respectively. Though this latter estimate is lower than conventionally acceptable levels (0.7), it is consistent with those generated from similar studies (Bell, Gitomer, McCaffrey, Hamre, & Pianta, 2012; Kane & Staiger, 2012).[1]

[1] Reliability estimates for the MQI from the MET study were lower. One reason for this may be that MET used the MQI Lite and not the full MQI instrument used in this study. The MQI Lite has raters provide only overarching dimension scores, while the full instrument asks raters to score teachers on up to five items before assessing an overall score. Another reason likely is related to differences in scoring designs. MET had raters score 30 minutes of instruction from

The CLASS instrument captures more general classroom quality. By design, the instrument is split into three dimensions. Based on factor analyses described above, I utilize two: *Classroom Emotional Support*, which focuses on the classroom climate and teachers' interactions with students; and *Classroom Organization*, including behavior management and productivity of the lesson. Following the protocol provided by instrument developers, one certified and trained rater watched and scored each lesson on 11 items for each fifteen-minute segment on a scale from Low (1) to High (7). I reverse code one item from the *Classroom Organization* dimension, "Negative Climate," to align with the valence of the other items. Therefore, in all cases, higher scores indicate better performance. Using the same method as above, I estimate reliabilities of 0.55 for *Classroom Emotional Support* and 0.65 for *Classroom Organization*.

In Table 2, I present summary statistics of teacher-level scores that are averaged across raters (for the MQI), segments, and lessons. For the MQI, mean scores are slightly lower than the middle of the scale itself: 1.26 for *Ambitious Mathematics Instruction* (out of 3; sd = 0.12) and 1.12 for *Mathematical Errors and Imprecisions* (out of 3; sd = 0.12). For the CLASS, mean scores are centered above the middle of the scale: 4.26 for *Classroom Emotional Support* (out of 7; sd = 0.55) and 6.52 for *Classroom Organization* (out of 7; sd = 0.44). Pairwise correlations between these teacher-level scores range from roughly zero (between *Mathematical Errors and Imprecisions* and the two dimensions on the CLASS instrument) to 0.44 between *Classroom*

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each lesson. Comparatively, in this study, raters provided scores for the whole lesson, which is in line with recommendations made by Hill, Charalambous, and Kraft (2012) in a formal generalizability study. Finally, given MET's intent to validate observation instruments for the purpose of new teacher evaluation systems, they utilized a set of raters similar to the school leaders and staff who will conduct these evaluations in practice. In contrast, other research shows that raters who are selectively recruited due to a background in mathematics or mathematics education and who complete initial training and ongoing calibration score more accurately on the MQI than those who are not selectively recruited (Hill, Charalambous, Blazar, et al., 2012).

Emotional Support and *Classroom Organization*. *Ambitious Mathematics Instruction* is more consistently related to the other instructional quality dimensions, with correlations between 0.19 and 0.34. These correlations are high enough to suggest that high-quality teachers who engage in one type of instructional practice may also engage in others, but not too high to indicate that dimensions measure the same construct.

Table 2

Univariate and Bivariate Descriptive Statistics of Instructional Quality Dimensions

	Univariate Statistics					Pairwise Correlations			
	Teacher Level		School-Grade-Year Level		Adjusted Intraclass Correlation	Ambitious Mathematics Instruction	Mathematical Errors and Imprecisions	Classroom Emotional Support	Classroom Organization
	Mean	SD	Mean	SD					
Ambitious Mathematics Instruction	1.26	0.12	1.27	0.10	0.69	1			
Linking and Connections									
Explanations									
Multiple Methods									
Generalizations									
Math Language									
Remediation of Student Difficulty									
Use of Student Productions									
Student Explanations									
Student Mathematical Questioning and Reasoning									
Enacted Task Cognitive Activation									
Mathematical Errors and Imprecisions	1.12	0.12	1.12	0.08	0.52	-0.33***	1		
Major Mathematical Errors									
Language Imprecisions									
Lack of Clarity									
Classroom Emotional Support	4.26	0.55	4.24	0.34	0.55	0.34***	-0.01	1	
Positive Climate									
Teacher Sensitivity									
Respect for Student Perspectives									
Classroom Organization	6.32	0.44	6.33	0.31	0.65	0.19***	0.05	0.44***	1
Negative Climate									
Behavior Management									
Productivity									

Notes: Statistics generated from all available data. MQI items (from *Ambitious Mathematics Instruction* and *Mathematical Errors and Imprecisions*) on a scale from 1 to 3. CLASS items (from *Classroom Emotional Support* and *Classroom Organization*) on a scale from 1 to 7.

As I discuss below, my identification strategy relies on instructional quality scores at the school-grade-year level. While this strategy loses between-teacher variation, which likely is the majority of the variation in instructional quality scores, I still find substantive variation in instructional quality scores within schools, across grades and years. In Table 3, I decompose the variation in school-grade-year scores into two components: the school-level component, which describes the percent of variation that lies across schools, and the residual component, which describes the rest of the variation that lies within schools. For all four instructional quality dimensions, I find that at least 40% of the variation in school-grade-year scores lies within schools. This leads me to conclude that there is substantive variation within schools at the school-grade-year level to exploit in this analysis.

Table 3
Variance Decomposition of School-Grade-Year Instructional Quality Scores

	School	Residual
Ambitious Mathematics Instruction	0.59	0.41
Mathematical Errors and Imprecisions	0.46	0.54
Classroom Emotional Support	0.45	0.55
Classroom Organization	0.52	0.48

Notes: Sample includes 76 school-grade-years.

In order to minimize noise in these observational measures, I use all available lessons for each teacher (Hill, Charalambous, & Kraft, 2012). Teachers who participated in the study for one year had three lessons, on average, while those who participated in the study for two years generally had six lessons. A second benefit of this approach is that reduces the possibility for bias due to unobserved classroom characteristics that affect both instructional quality and student

outcomes (Kane, Taylor, Tyler, & Wooten, 2011).[2] This is because, in roughly half of cases, scores represent elements of teachers' instruction from the prior year or future year, in addition to the current year. Specifically, I utilize empirical Bayes estimation to shrink scores back toward the mean based on their precision (see Raudenbush & Bryk, 2002). To do so, I specify the following hierarchical linear model using all available data, including teachers beyond my identification sample:

$$(1) \qquad OBSERVATION_{lj} = \mu_j + \varepsilon_{lj}$$

where the outcome is the observation score for lesson *l* and teacher *j*, μ_j is a random effect for each teacher *j*, and ε_{lj} is the error term. I utilize standardized estimates of the teacher-level random effect as each teacher's observation score. Most distributions of these variables are roughly normal. For identification, I average these scores within each school-grade-year. I do not re-standardize these school-grade-year scores in order to interpret estimates in teacher-level standard deviation units, which are more meaningful than school-grade-year units.

3.2.2 Student Demographic and Test-Score Data

[2] Kane et al. (2011) argue that cotemporaneous measurement of teacher observation scores and student outcomes may bias estimates due to class characteristics that affect both the predictor and the outcome. I do not do so here for both practical and substantive reasons. The sample of school-grade-years in which all teachers have out-of-year observation scores is too limited to conduct the same sort of analysis. In addition, as this study is interested in the effect of instruction on student outcomes, I want to utilize scores that capture the types of practices and activities in which students themselves are engaged.

At the same time, I am able to examine the extent to which Kane et al.'s hypothesis plays out in my own data. To do so, I explore whether changes in classroom composition predict changes in instructional quality for those 45 teachers for whom I have two years of observation data. In Appendix Table A1, I present estimates from models that regress each instructional quality dimension on a vector of observable class characteristics and teacher fixed effects. Here, I observe that classroom composition only predicts within-teacher, cross-year differences in *Classroom Emotional Support* ($F = 2.219$, $p = 0.035$). This suggests that attention to omitted variables related both to *Classroom Emotional Support* and student achievement may be important.

One source of student-level data is district administrative records. Demographic data include gender, race/ethnicity, special education (SPED) status, limited English proficiency (LEP) status, and free- or reduced-price lunch (FRPL) eligibility. I also utilize prior-year test scores on state assessments in both math and reading, which are standardized within district by grade, subject, and year using the entire sample of students in each district, grade, subject, and year.

Student outcomes were measured in both fall and spring on a new assessment developed by researchers who created the MQI in conjunction with the Educational Testing Service (see Hickman, Fu, & Hill, 2012). Validity evidence indicates internal consistency reliability of 0.82 or higher for each form across the relevant grade levels and school years. Three key features of this test make it ideal for this study. First, the test is common across all districts and students in the sample, which is important given evidence on the sensitivity of statistical models of teacher effectiveness to different achievement tests (Lockwood, McCaffrey, Hamilton, Stecher, Le, & Martinez, 2007; Papay, 2011). Second, the test is vertically aligned, allowing me to compare achievement scores for students in fourth versus fifth grade. Third, the assessment is a relatively cognitively demanding test, thereby well aligned to many of the teacher-level practices assessed in this study, particularly those captured on the MQI instrument. It likely also is similar to new mathematics assessments administered under the Common Core (National Governors Association for Best Practices, 2010). Lynch, Chin, and Blazar (2015) coded items from this assessment for format and cognitive demand using the *Surveys of Enacted Curriculum* framework (Porter, 2002). They found that the assessment often asked students to solve non-routine problems, including looking for patterns and explaining their reasoning. Roughly 20% of items required short responses.

3.2.3 Teacher Survey

Information on teachers' background, knowledge, and skills were captured on a teacher questionnaire administered in the fall of each year. Survey items about teachers' background include whether or not the teacher earned a bachelor's degree in education, amount of undergraduate or graduate coursework in math and math courses for teaching (2 items scored from 1 [No Classes] to 4 [Six or More Classes], internal consistency reliability (α) = 0.66), route to certification, and whether or not the teacher had a master's degree (in any subject). Relatedly, the survey also asked about the number of years of teaching experience in math.

Next, I capture teachers' knowledge of content and of their students. Teachers' content knowledge was assessed on items from both the Mathematical Knowledge for Teaching assessment (Hill, Schilling, & Ball, 2004) and the Massachusetts Test for Educator Licensure. Teacher scores were generated by IRTPro software and were standardized in these models using all available teachers, with a reliability of 0.92. Second are scores from a test of teachers' knowledge of student performance. These scores were generated by providing teachers with student test items, asking them to predict the percent of students who would answer each item correctly, then calculating the distance between each teacher's estimate and the actual percent of students in their class who got each item correct. Similar to instructional quality scores, I report reliability as adjusted intraclass correlations, which are 0.71 and 0.74 for grades four and five, respectively. To arrive at a final scale, I averaged across items and standardized.

Finally, two items refer to additional classroom behaviors that aim to increase student achievement. The first is teachers' preparation for class, which asks about the amount of time each week that teachers devoted to out-of-class activities such as grading, preparing lesson materials, reviewing the content of the lesson, and talking with parents (4 items scored from 1

[No Time] to 5 [More than Six Hours], $\alpha = 0.84$). The second construct is formative assessment, which asks how often teachers evaluated student work and provided feedback (5 items scored from 1 [Never] to 5 [Daily or Almost Daily], $\alpha = 0.74$).[3]

In Table 4, I present correlations between these characteristics and the four instructional quality dimensions. The strongest correlation is between *Mathematical Errors and Imprecisions* and mathematical content knowledge ($r = -0.46$). This suggests that teachers' knowledge of the content area is moderately to strongly related to their ability to present correct material in class. The sign of this relationship is correct, in that higher scores on *Mathematical Errors and Imprecisions* means that more errors are made in instruction, while higher scores on the content knowledge test indicate stronger understanding of math. Content knowledge also is related to *Ambitious Mathematics Instruction* ($r = 0.26$). Interestingly, math coursework is related to *Classroom Organization,* and *Mathematical Errors and Imprecisions* is related to formative assessment ($r = 0.24$), even though these constructs are not theoretically related. Together, this suggests that the dimensions of instructional quality generally are distinct from other measures often used as a proxy for teacher or teaching quality.

Table 4
Correlations Between Teacher Practices, Skills, and Background Characteristics

	Ambitious Mathematics Instruction	Mathematical Errors and Imprecisions	Classroom Emotional Support	Classroom Organization
Bachelor's Degree in Education	-0.14	-0.03	-0.07	0.13
Math Coursework	0.08	0.08	0.15	0.30***
Master's Degree	0.10	-0.05	0.00	-0.12
Traditional Certification	0.09	-0.17~	0.12	0.12
Experience	-0.07	0.15	-0.04	0.05
Mathematical Content Knowledge	0.26**	-0.46***	0.03	0.01
Knowledge of Student Performance	0.18~	-0.16	0.00	0.09
Preparation for Class	0.02	0.07	-0.04	0.10
Formative Assessment	-0.01	0.24**	0.14	0.17~

Notes: ~ p<0.10, * p<0.05, ** p<0.01, ***p<0.001

[3] Between three and six teachers are missing data for each of these constructs. Given that these data are used for descriptive purposes and as controls, in these instances I impute the mean value for the district. For more information on these scales, see Hill, Blazar, and Lynch (2015).

4. Identification Strategy and Tests of Assumptions

In order to estimate the relationship between high-quality instruction and students' mathematics achievement, my identification strategy must address two main challenges: non-random sorting of students to teachers and omitted measures of teachers' skills and practices. I focus on each in turn.

4.1 Non-Random Sorting of Students to Teachers

Non-random sorting of students to teachers consists of two possible components: the sorting of students to schools and of students to classes or teachers within schools. In Table 5, I explore the extent to which these types of sorting might bias results by regressing baseline test scores on all four dimensions of instructional quality (see Kane et al., 2011). Comparing teachers within districts, *Ambitious Mathematics Instruction* is positively related to baseline achievement. This suggests, unsurprisingly, that teachers with higher-quality math instruction tend to be assigned to higher-achieving students. Interestingly, though, only part of this relationship is explained by differences in instructional quality and student achievement across schools. Comparing teachers within schools, the magnitude of the relationship between *Ambitious Mathematics Instruction* and baseline achievement is substantively smaller but still statistically significant. Further, I now observe a positive relationship between *Classroom Organization* and baseline test scores. This indicates that within-school sorting and the matching of students to teachers may occur differently than across-school sorting but that it likely serves as an additional source of bias.

Table 5

Relationships Between Assigned Students' Incoming Achievement and Instructional Quality

	Within Districts	Within Schools
Ambitious Mathematics Instruction	0.180***	0.060*
	(0.026)	(0.028)
Mathematical Errors and Imprecisions	-0.022	-0.034
	(0.021)	(0.022)
Classroom Emotional Support	-0.013	-0.018
	(0.018)	(0.023)
Classroom Organization	-0.003	0.087**
	(0.024)	(0.029)

Notes: ~ p< .10, * p<.05, ** p<.01, ***p<.001. Columns contain estimates from separate regressions. Robust standard errors in parentheses. All models control for district-by-grade-by-year fixed effects. Sample includes 3,203 students, 111 teachers, and 76 school-grade-years.

In light of non-random sorting, I begin by specifying models that control for a host of observable student and class characteristics, including prior achievement. Further, following Kane, Taylor, Tyler, and Wooten (2011), I include school fixed effects to account for unobserved differences across schools, other than instructional quality, that also affect student achievement. Finally, to address sorting of students to classes or teachers within schools, I exploit an important logistical and structural constraint of schools – that students may be sorted within but not across grades and years. This is because, in most cases, students advance with a given cohort from one grade to the next. Therefore, similar to Rivkin, Hanushek, and Kain (2005), I exploit between-cohort differences by aggregating teachers' observation scores to the school-grade-year level. They argue that "aggregation to the grade level circumvents any problems resulting from classroom assignment" (p. 426). Doing so restricts identifying variation to that observed across grades – e.g., between fourth-grade teachers in one year and fifth-grade teachers in the same,

following, or former school year. In a few instances where grade-level composition changes from one year to the next, there also is identifying variation between the set of fourth-grade teachers in one year and the set of fourth-grade teachers in the following or former school year, and similarly for fifth-grade teachers in one year and fifth-grade teachers in another year

The hypothesized model that describes this relationship is outlined in equation 2:

$$A_{idsgcjt} = \beta \overline{OBSERVATION}_{sgt} + \zeta\left(f\left(A_{it-1}\right)\right) + \pi X_i + \varphi \overline{X}_{cjt} + \sigma_{dgt} + \theta_s + \varepsilon_{idsgcjt} \qquad (2)$$

where $A_{idsgcjt}$is the end-of-year test score for student *i* in district *d*, school *s*, grade *g*, and class *c* with teacher *j* at time *t*; $\overline{OBSERVATION}_{sgt}$ is a vector of instructional quality scores that are averaged across teachers within each school-grade-year; $f\left(A_{it-1}\right)$ is a cubic function of prior achievement on the fall baseline assessment, as well as on the prior-year state assessments in both math and reading; X_i is a vector of observable student-level characteristics; $\overline{X}_{cjt}$ aggregates these and prior achievement measures to the class level. I include district-by-grade-by-year fixed effects, σ_{dgt}, to account for differences in the scaling of state standardized test scores. As discussed above, I also include fixed effects for schools, θ_s, as part of my identification strategy. I calculate standard errors that are clustered at the school-grade-year level to account for heteroskedasticity in the student-level errors, $\varepsilon_{idsgcjt}$, and non-zero covariance among those students attending the same school in the same grade and year (Kane, Rockoff, & Staiger, 2008).

The key identifying assumption of this model is that within-school, between-grade, and cross-cohort differences in average instructional quality scores are exogenous (see Woessmann & West, 2006 for a discussion of this assumption and strategy as it pertains to class size). While the validity of this assumption is difficult to test directly, I can examine ways that it may play out in practice. In particular, this assumption would be violated by strategic grade assignments in which teachers are shifted across grades due to a particularly strong or weak incoming class, or

where students are held back or advanced an additional grade in order to be matched to a specific teacher.

Although these practices are possible in theory, I present evidence that such behavior does not threaten inferences about variation in instructional quality scores. I do observe that 30 teachers were newly assigned to their grade, either because they switched from a different grade in the prior year (before joining the study) or because they moved into the district. In Table 6, I examine differences between switchers and non-switchers on observable characteristics within school-year cells. In addition to comparing teachers on the characteristics listed in Tables 1 and 2, I include average scores on all three baseline achievement tests; I also include state value-added scores in math.[4] Here, I find that switchers have students with lower prior-year achievement on state math and reading exams (p = 0.037 and 0.002, respectively). Importantly, though, there are no differences between switchers and non-switchers on any of the observational rubric dimensions, any of the teacher survey constructs, or state value-added scores. Nor can I detect differences between these two groups when all observable traits are tested jointly (F = 1.159, p = 0.315).[5] This suggests that, even though switchers tend to have lower-achieving students, they are unlikely to be matched to these classes based on observed quality. With regard to sorting of students to grade, fewer than 20 were retained from the previous year or skipped a grade. I drop these from the analytic sample.

[4] Value-added scores are calculated from a model similar to equation (2). Here, I regress end-of-year student mathematics test scores on state assessments on a vector of prior achievement; student-, class-, and school-level covariates; and district-by-grade-by-year fixed effects. I predict a teacher-level random effect as the value-added score. I utilize all years of data and all teachers in the sample districts and grades to increase the precision of my estimates (Goldhaber & Hansen, 2012; Koedel & Betts 2011; Schochet & Chiang, 2013).

[5] In some instances, mean scores for both switchers and non-switchers on standardized variables fall below or above zero (e.g., *Classroom Emotional Support*). This is possible given that variables were standardized across all teachers in the study, not just those in the identification sample.

Table 6
Differences Between Teachers Who Switch Grade Assignments and Those Who Do Not

	Switchers	Non-Switchers	*p*-value on Difference
Instructional Quality Dimensions			
Ambitious Mathematics Instruction	-0.05	0.03	0.660
Mathematical Errors and Imprecisions	-0.07	-0.20	0.463
Classroom Emotional Support	-0.18	-0.25	0.752
Classroom Organization	-0.22	-0.11	0.596
Other Measures of Teacher Quality			
Bachelor's Degree in Education	63.0	42.7	0.169
Math Coursework	2.2	2.4	0.259
Master's Degree	74.4	77.4	0.781
Traditional Certification	69.7	74.7	0.613
Experience	7.8	10.1	0.208
Mathematical Content Knowledge	-0.19	-0.01	0.558
Knowledge of Student Performance	0.20	0.06	0.519
Preparation for Class	3.3	3.3	0.981
Formative Assessment	3.5	3.7	0.318
Student Achievement Measures			
Fall Project-Administered Math Test	-0.35	-0.12	0.318
Prior-Year State Math Test	-0.05	0.08	0.037
Prior-Year State Reading Test	-0.09	0.10	0.002
State Value-Added in Math	-0.03	-0.01	0.646
Join Test		*F*-statistic	1.098
		p-value	0.367
Teacher-Year Observations	30	126	

Notes: Means and *p*-values estimated from individual regressions that control for school-year, which is absorbed in the model. See Table 1 for scale of teacher quality measures. All other items are standardized.

A second assumption underlying the logic of this strategy is that identification holds only when all teachers at a given school-grade-year are in the study. If only a portion of the teachers participate, then there may be bias due to the selection of students assigned to these teachers. To address this concern, I limit my final analytic sample to school-grade-years in which I have full participation of teachers. I am able to identify these teachers as I have access to class rosters for all teachers who work in the sample districts. I exclude from these school-grade-year teams teachers who teach self-contained special education or bilingual classes, as the general population of students would not be sorted to these teachers' classes.[6]

[6] I identify these specialized classes in cases where more than 50% of students have this designation.

By dropping certain school-grade-year observations, I limit the sample from which I am able to generalize results. In this sense, I compromise external validity for internal validity. However, below I discuss the comparability of teachers and school-grade-years included in my identification sample to those that I exclude either because they did not participate in data collection through the NCTE project or because they did not meet the sample conditions I describe above.

4.2 Omitted Variables Bias

Given non-random sorting of instructional quality to teachers, estimating the effect of these practices on mathematics achievement also requires isolating them from other characteristics that are related both to observation rubric scores and to student test scores. I focus on characteristics that prior research suggests may fit the definition of omitted variables bias in this type of analysis.

Review of prior research indicates that several observable characteristics are related both to student achievement and instructional quality. Studies indicate that students experience larger test score gains in math from teachers with prior education and coursework in this content area (Boyd, Grossman, Lankford, Loeb, & Wyckoff, 2009; Wayne & Youngs, 2003), some forms of alternative certification such as Teach for America relative to traditional certification (Clark et al, 2013; Decker, Mayer, & Glazerman, 2004), more experience in the classroom (Chetty et al., 2011; Papay & Kraft, forthcoming; Rockoff, 2004), and stronger content knowledge (Metzler & Woessmann, 2012). Emerging work also highlights the possible role of additional professional competencies, such as knowledge of student performance, in raising student achievement (Kunter, Klusmann, Baumert, Richter, Voss, & Hachfeld, 2013; Sadler, Sonnert, Coyle, Cook-Smith, & Miller, 2013). These factors also appear to predict some dimensions of instructional

quality in this or other datasets (see Table 3 and Hill, Blazar, & Lynch, 2015 for further discussion).

Because it is possible that I am missing other important characteristics – namely unobservable ones – I test the sensitivity of results to models that include different sets of teacher-level covariates. I also interpret results cautiously. Despite this limitation, I believe that my ability to isolate instructional practices from a range of other teacher traits and skills is an advance beyond similar studies.

5. Results

5.1 Main Results

In Table 7a, I present models examining the relationship between instructional quality and student achievement. This first set of models examines the robustness of estimates to specifications that attempt to account for the non-random sorting of students to schools and teachers. I begin with a basic model (Model A) that regresses students' spring test score on teacher-level observation scores. I include a cubic function of fall/prior achievement on the project-administered test and state standardized tests in math and reading; utilizing all three tests of prior achievement allows me to compare students with similar scores on low- and high-stakes tests across both subjects, increasing the precision of my estimates. I also include district-by-grade-by-year dummy variables to account for differences in scaling of tests; and vectors of student-, class-, and school-level covariates. Next, I replace school-level covariates with school fixed effects (Model B). In Model C, I retain the school fixed effects and replace observation scores at the teacher level with those at the school-grade-year level. This model matches equation (2) above. Finally, in order to ensure that school-specific year effects do not drive results, I replace school fixed effects with school-by-year fixed effects in Models D. For all

models, I limit the sample to those school-grade-years where all teachers from participating school-grades-years are in the study. Robust standard errors clustered at the school-grade-year level are reported in parentheses.[7]

Table 7a

Relationships Between Students' Mathematics Achievement and Instructional Quality, Accounting for Non-Random Sorting

	Model A	Model B	Model C	Model D
Ambitious Mathematics Instruction	0.061	0.095*	0.097*	0.109*
	(0.038)	(0.037)	(0.042)	(0.052)
Mathematical Errors and Imprecisions	-0.033	-0.040~	-0.050~	-0.053~
	(0.022)	(0.023)	(0.026)	(0.029)
Classroom Emotional Support	-0.028	-0.001	-0.032	-0.026
	(0.021)	(0.023)	(0.035)	(0.037)
Classroom Organization	0.026	-0.002	-0.003	-0.015
	(0.025)	(0.024)	(0.034)	(0.037)
Student Covariates	X	X	X	X
Class Covariates	X	X	X	X
District-by-Grade-by-Year Fixed Effects	X	X	X	X
School Covariates	X			
School Fixed Effects		X	X	
Instructional Quality at School-Grade-Year Level			X	X
School-by-Year Fixed Effects				X

Notes: ~ p<0.10, * p<0.05, ** p<0.01, ***p<0.001. Columns contain estimates from separate regressions. Robust standard errors clustered at the school-grade-year level in parentheses. Sample includes 3,203 students, 111 teachers, and 76 school-grade-years.

In Model C, intended to account for non-random sorting of students to schools and teachers, I find that instructional quality dimensions focused on the mathematics presented in the classroom are related to students' math achievement. Specifically, I find a statistically significant and positive coefficient for *Ambitious Mathematics Instruction* of 0.10 sd; the coefficient for *Mathematical Errors and Imprecisions* of -0.05 sd is marginally significant.

Interestingly, these estimates are larger in magnitude than those from Models A and B. Comparison of estimates to Model A implies that schools and/or classrooms where instruction is higher quality tend to have below-average test-score growth. The fact that estimates in Model C

[7] I also test the robustness of results to clustering of standard errors at the school-year level, and find that standard errors and significance levels presented below do not change substantively.

are larger than those in Model B is surprising. By limiting variation to school-grade-years, I expected to calculate lower-bound estimates of the relationship between instructional quality and student achievement (see Rivkin, Hanushek, & Kain, 2005). One possible explanation for my findings may be that school-grade-year scores are picking up the quality of teaching teams, which also is related to student achievement. At the same time, these differences are not large. Further, standard errors are larger in Model C than in Model B, as I would expect given more limited variation in my main predictor variables. Finally, I find that estimates in Model D, which replace school fixed effects with school-by-year fixed effects, are similar in magnitude to those in Model C. This indicates that year effects do not drive results. As before, standard errors are larger than those in Model C given more limited identifying variation. I find no statistically significant relationships for the two other dimensions of instruction.

In Table 7b, I re-estimate results from Model C controlling for different sets of teacher characteristics. I focus on four categories of covariates: education and certification (Model E), teaching experience (Model F), knowledge (Model G), and non-instructional classroom behaviors (Model H). In Model I, I include all four sets of predictors. Similar to instructional quality dimensions, these covariates are averaged to the school-grade-year level. Here, I find that estimates for *Ambitious Mathematics Instruction* are fairly robust to inclusion of these control variables. In Model G, which controls for two measures of teacher knowledge, I find a marginally significant estimate of 0.08 sd. This slight attenuation makes sense given the positive relationship between mathematical content knowledge and *Ambitious Mathematics Instruction* noted earlier. Interestingly, coefficients from models that include other sets of covariates are slightly larger than my estimate of 0.10 sd from Model C; in Model I, which controls for all teacher characteristics, the resulting estimate is roughly 0.11 sd. One reason for this may be that

be these additional predictors are negatively related either to instructional quality or to student achievement. Earlier, I showed a negative, though not statistically significant, correlation between *Ambitious Mathematics Instruction* and bachelor's degree in education; here, I observe small but negative relationships to student achievement for bachelor's degree in education, math coursework, traditional certification, and preparation for class. I am cautious in placing too much emphasis on these differences, as they are not large. However, these patterns suggest that some omitted variables may lead to upward bias while others lead to downward bias.

Table 7b

Relationships Between Students' Mathematics Achievement and Instructional Quality, Accounting for Possible "Omitted" Variables

	Model E	Model F	Model G	Model H	Model I
Ambitious Mathematics Instruction	0.124**	0.096*	0.083~	0.121**	0.114*
	(0.042)	(0.039)	(0.045)	(0.041)	(0.044)
Mathematical Errors and Imprecisions	-0.049~	-0.049~	-0.035	-0.038	-0.028
	(0.027)	(0.029)	(0.026)	(0.027)	(0.035)
Classroom Emotional Support	-0.038	-0.031	-0.025	-0.044	-0.041
	(0.031)	(0.036)	(0.036)	(0.034)	(0.036)
Classroom Organization	0.010	-0.002	-0.009	-0.002	-0.002
	(0.035)	(0.033)	(0.034)	(0.035)	(0.039)
Bachelor's Degree in Education	0.010				-0.004
	(0.065)				(0.072)
Math Coursework	-0.027				-0.019
	(0.021)				(0.028)
Master's Degree	0.086				0.022
	(0.070)				(0.075)
Traditional Certification	-0.013				-0.019
	(0.068)				(0.077)
Experience		-0.001			-0.000
		(0.004)			(0.005)
Mathematical Content Knowledge			0.017		0.008
			(0.020)		(0.031)
Knowledge of Student Performance			0.035		0.038
			(0.041)		(0.044)
Preparation for Class				-0.054~	-0.044
				(0.030)	(0.038)
Formative Assessment				0.028	0.027
				(0.032)	(0.037)

Notes: ~ p< .10, * p<.05, ** p<.01, ***p<.001. Columns contain estimates from separate regressions. Robust standard errors clustered at the school-grade-year level in parentheses. All models control for student and class covariates, as well as district-by-grade-by-year and school fixed effects. Instructional quality and background characteristics are averaged at the school-grade-year level. Sample includes 3,203 students, 111 teachers, and 76 school-grade-years.

The relationship between *Mathematical Errors and Imprecisions* and student achievement is more sensitive to inclusion of control variables. Original estimates from Model C are attenuated most significantly when controlling for teachers' mathematical content knowledge; the resulting estimate of roughly -0.04 sd in Model G is no longer marginally statistically significant. This attenuation is unsurprising given a moderate to strong relationship between *Mathematical Errors and Imprecisions* and mathematical content knowledge noted earlier (r = -0.46). Therefore, it is difficult to tell whether student achievement is negatively impacted by teachers' lack of content knowledge, the way that this lack of knowledge leads to errors and imprecisions in the presentation of material, or a related construct. When I include all sets of predictors in the same model (Model I), the estimate for *Mathematical Errors and Imprecisions* is -0.03 sd and not statistically significant.

5.2 Generalizability of Results Beyond Identification Sample

Finally, in Table 8, I examine whether teachers and schools included in my identification sample are representative of those in their respective districts. Because I do not have instructional quality scores for all district teachers, for this analysis I draw on mathematics value-added scores using state assessment data. I also compare observable characteristics of school-grade-years from my identification sample to those across the rest of the sample districts, looking for differences on each characteristic individually and as a group. *P*-values testing the difference between sample means are calculated through a regression framework that controls for district, as recruitment of schools and teachers occurred at this level. In both cases of teachers and school-grade-years, I cannot reject the null hypothesis that my identification sample is the same as the rest of the district populations (for differences in teachers' value-added scores: p = 0.123;

for joint differences in observable characteristics of school-grade-years: $F = 0.902$, $p = 0.531$). Therefore, I conclude that results likely generalizable to these populations.

Table 8

Differences Between Identification Sample and District Populations

	In Identification Sample	Out of Identification Sample	*p*-value on Difference
Teacher Characteristic			
State Value-Added	-0.02	0.00	0.123
Teacher-Year Observations	156	1,334	
School-Grade-Year Characteristics			
Male	49.1	50.1	0.361
African-American	53.7	55.3	0.659
Asian	4.6	3.9	0.404
Hispanic	26.6	26.0	0.833
White	11.6	11.6	0.996
FRPL	74.2	76.3	0.504
SPED	17.1	15.7	0.240
LEP	21.3	20.8	0.810
Prior-Year State Math Test	-0.02	0.04	0.299
Prior-Year State Reading Test	0.00	0.05	0.409
Joint Test		*F*-statistic	0.902
		p-value	0.531
School-Grade-Year Observations	76	511	

Notes: Means and *p*-values calculated from individual regressions that control for district. School-grade-year demographic characteristics are percents; test scores are standardized.

6. Discussion and Conclusion

This study provides some of the strongest evidence to date on the relationship between specific instructional dimensions and students' mathematics achievement. Like others (e.g., Kane et al., 2013; Kane & Staiger, 2012; Kane et al., 2011), I utilize observation instruments that capture instructional quality within teachers' own classrooms. I also draw on established econometric methods to account for the non-random sorting of students to teachers (e.g., Rivkin, Hanushek, & Kain, 2005). Importantly, I build on past work by examining multiple dimensions of teaching practice, including content-specific elements of instruction and more general pedagogical strategies. Further, I examine the sensitivity of results to models that control for different sets of teacher characteristics. This allows me to isolate dimensions of instructional

quality from the most likely observable characteristics that might threaten the internal validity of my results. To my knowledge, no other studies are able to control for this broad set of teaching practices and teacher characteristics. While it is possible that estimates are sensitive to other observed or unobserved characteristics not included in these data, my findings provide strong suggestive evidence of teaching dimensions that support student achievement.

Results indicate that inquiry-oriented instruction is positively related to student outcomes on a low-stakes math test, with an effect size of roughly 0.10 sd. This finding lends support to decades worth of reform to refocus mathematics instruction toward inquiry and concept-based teaching (National Council of Teachers of Mathematics, 1989, 1991, 2000), as well as positive results of some of these types of activities in laboratory settings (e.g., Star & Rittle-Johnson, 2009). In some analyses, I also find smaller effect sizes for incorrect presentation of content, though estimates are sensitive to the set of covariates included in the model, particularly teachers' content knowledge. At the same time, even the smallest estimate of roughly 0.03 sd (see Model I in Table 7b) is similar in magnitude to estimates of the relationship between mentor evaluations and student achievement (Rockoff & Speroni, 2010), suggesting that this finding may still be substantively significant.

Finally, I find no relationship between classroom climate or classroom management and student achievement. These results diverge from recent research highlighting the importance of classroom organization and interactions with students, often above other classroom features (Grossman, Loeb, Cohen, & Wyckoff, 2013; Stronge, Ward, & Grant, 2011). In particular, Kane and co-authors (2011, 2012, 2013) found positive relationships between these sorts of classroom practices, as captured on the Framework for Teaching observation instrument, and student achievement; estimates were similar in magnitude to the relationship I find between *Ambitious*

Mathematics Instruction and student outcomes. One reason for these differences may be that these other studies did not account for additional dimensions of teacher and teaching quality. Therefore, the observed relationship between classroom organization and student achievement may be driven by other practices and skills that are related to this type of instruction. Another reason may be that the outcome used to measure math achievement in this study is a low-stakes test that emphasizes cognitively demanding mathematics practices. Classroom organization and interactions with students may in fact be important contributors to high-stakes achievement tests or non-cognitive outcomes. This is an important topic for future research.

Evidence on the relationship between specific types of teaching and student achievement raises the question of how to get more teachers who engage in these practices into classrooms. Following Murnane and Cohen (1986), I argue that incentives are unlikely to prove effective here, as teachers may not know *how* to improve their instruction. Therefore, I propose two possible pathways. First, an array of recent literature highlights the potential use of observation instruments themselves to remediate teacher practice. Despite mixed results on the effect of standard professional development programs on teachers' content knowledge, instructional practices, or student achievement (Garet et al., 2011; Yoon, Duncan, Lee, Scarloss, & Shapley, 2007), new experimental studies highlight positive effects of more intensive coaching programs that utilize observation instruments to improve teacher behaviors and, in some cases, student outcomes (Allen et al., 2011; Blazar & Kraft, forthcoming; McCollum, Hemmeter, & Hsieh, 2011; Taylor & Tyler, 2012). Thus far, this sort of work has focused on use of observation instruments to capture general teaching practices and those specific to literacy instruction. However, it is possible that findings also extend to inquiry-oriented practices in mathematics.

A second pathway to increase the quality of classroom teaching may also focus on selective recruitment of teachers with content-area expertise. My findings show a moderate to strong relationship between teachers' knowledge of math and the way that this content is enacted in the classroom. Further, I find suggestive evidence of a relationship between incorrect presentation of content and student outcomes. While more research is needed to confirm these relationships, these patterns may inform processes by which education preparation programs and state licensing agencies screen prospective elementary math teachers. A survey of degree pathways indicates minimal requirements for entry and a high degree of variability in the type of training pre-service teachers receive in mathematics. In addition, in all but a few states, elementary teachers can pass their licensing exam without passing the math sub-section (Epstein & Miller, 2011). It is possible that creating more stringent requirements into the workforce related to math knowledge could lead to more accurate and precise presentation of content and to better student outcomes.

Filling elementary classrooms with teachers who engage in effective mathematics teaching practices will take time. Doing so likely will entail a variety of efforts, including improvements in professional development offerings that engage teachers substantively around their own teaching practices and stronger efforts to hire teachers with deep knowledge of mathematics. Importantly, though, the education community is beginning to gain an understanding of the types of teaching that contribute to student achievement.

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Appendix

Table A1
Relationships Between Instructional Quality and Class Composition

	Ambitious Mathematics Instruction	Mathematical Errors and Imprecisions	Classroom Emotional Support	Classroom Organization
Class Size	-0.069	0.020	-0.114	-0.029
	(0.069)	(0.059)	(0.077)	(0.061)
Male	0.016	-0.013	-0.002	-0.021
	(0.012)	(0.013)	(0.014)	(0.016)
African American	0.005	0.005	-0.038	0.022
	(0.023)	(0.026)	(0.034)	(0.029)
Asian	-0.015	-0.016	-0.037	0.060
	(0.037)	(0.038)	(0.052)	(0.039)
Hispanic	0.002	0.003	-0.036	0.030
	(0.022)	(0.024)	(0.034)	(0.026)
White	-0.017	0.012	0.005	0.035
	(0.035)	(0.035)	(0.043)	(0.036)
FRPL	-0.014	0.000	0.012	0.016
	(0.011)	(0.013)	(0.013)	(0.011)
SPED	-0.009	0.006	-0.035*	-0.018
	(0.010)	(0.012)	(0.013)	(0.012)
LEP	-0.003	0.004	0.004	0.014
	(0.010)	(0.017)	(0.018)	(0.019)
Fall Project-Administered Math Test	0.439	1.739	-2.384*	0.085
	(0.666)	(1.090)	(0.880)	(0.859)
Prior-Year State Math Test	-0.005	0.099	-0.984	-0.523
	(0.630)	(0.834)	(0.877)	(1.028)
Prior-Year State Reading Test	0.475*	-0.401	1.186**	-0.366
	(0.224)	(0.462)	(0.368)	(0.421)
Joint Test				
F-statistic	1.652	0.580	2.219	1.624
p-value	0.125	0.842	0.035	0.133

Notes: ~ p<0.10, * p<0.05, ** p<0.01, ***p<0.001. Columns contain estimates from separate regressions. Robust standard errors clustered at the school-grade-year level in parentheses. All models include teacher fixed effects. Sample includes 45 teachers who were in the study for two years.